Stage 8 Science — Practice Test 2

Paper 1: 45 minutes · calculator allowed · Paper 2: 45 minutes · Total 100 marks

Answer all questions. Write your answers in the spaces provided. Show your working where there is space for it. The number of marks for each question is shown in brackets [ ].

Name _____ Class _____ Date _____ Mark: ____ / 100

Abdelkriem Academy · practice paper written in the style of Cambridge Checkpoint. Not an official Cambridge paper.

Paper 1 — 45 minutes — 50 marks

1 Hassan feels tired and gets out of breath easily. At the hospital, a nurse takes a sample of his blood for a blood test. The table shows two of his results. **[1]**

Component of blood	Hassan's result	Normal range for an adult man
red blood cells (millions per mm^3)	3.6	4.5 to 5.9
white blood cells (per mm^3)	15 000	4 000 to 11 000

(a) Which component of Hassan's blood is **lower** than the normal range?

(b) Use your answer to (a) to explain why Hassan feels tired and gets out of breath. **[2]**

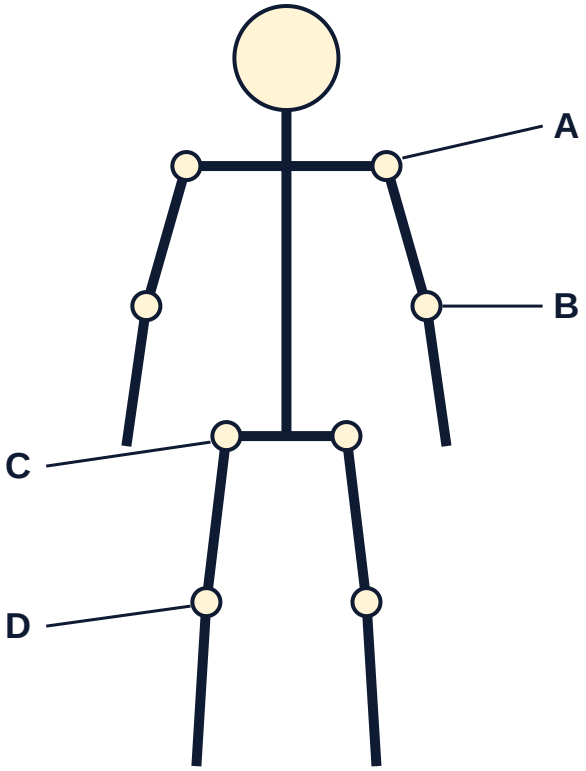
(c) Hassan's white blood cell count is **higher** than normal. What does this most likely show? **[1]**

- ☐ A) He is not eating enough iron.
- ☐ B) His plasma is carrying too much carbon dioxide.
- ☐ C) His body is fighting an infection caused by pathogens.
- ☐ D) His lungs are not taking in enough air.

(d) The liquid part of the blood is called plasma. Which list shows things that plasma transports? **[1]**

- ☐ A) blood cells, nutrients and carbon dioxide
- ☐ B) air, bones and pathogens only
- ☐ C) oxygen joined to haemoglobin only
- ☐ D) white blood cells only

2 After a fall, Hassan has an X-ray. The diagram is a simple model of the human skeleton. A, B, C and D are joints. **[1]**



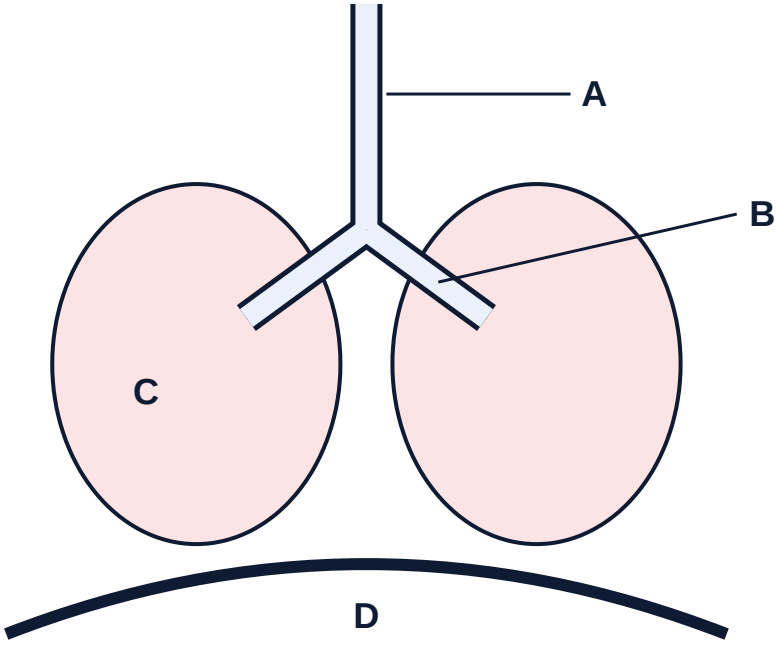
(a) Which **two** letters show **hinge** joints?

- ☐ A) A and C
- ☐ B) B and D
- ☐ C) A and B
- ☐ D) C and D

(b) Hassan's physiotherapist asks him to **straighten** his arm at joint B. Name the muscle that **contracts** to straighten the arm. **[1]**

(c) Explain why the muscles that move the arm at joint B must work as an **antagonistic pair**. **[2]**

3 In the hospital, a doctor uses this diagram to explain the human respiratory system to Hassan. **[1]**



(a) Which letter shows the **diaphragm**?

(b) Hassan breathes **in**. Which statement describes what happens? **[1]**

- ☐ A) The diaphragm relaxes and moves up, and the volume of the chest decreases.
- ☐ B) The intercostal muscles relax, so the ribs move up and out.
- ☐ C) The diaphragm contracts and moves up, so air is pushed out.
- ☐ D) The diaphragm contracts and flattens, and the volume of the chest increases.

(c) Oxygen moves from the air in the alveoli into the blood. Describe how. **[2]**

(d) Hassan says, "Breathing and respiration are the same thing." Which statement shows that he is **wrong**? **[1]**

- ☐ A) Breathing happens only in animals; respiration happens only in plants.
- ☐ B) Breathing moves air into and out of the lungs; respiration is a reaction in cells that releases energy.
- ☐ C) Breathing releases energy from glucose; respiration moves air into the lungs.
- ☐ D) Breathing uses carbon dioxide; respiration makes oxygen.

4 A food factory makes a breakfast cereal. This is part of the nutrition label. **[1]**

Nutrient	Amount in 100 g of cereal
energy	1 600 kJ
protein	9 g
carbohydrate	76 g
fat	3 g
calcium	0.5 g
iron	0.012 g
vitamin D	added

(a) Which nutrient in the table gives this cereal **most** of its energy?

(b) Ali eats a 40 g bowl of the cereal. Calculate the energy in his bowl of cereal. Show your working. **[2]**

Working

Answer ____ kJ

(c) Vitamin D is added to the cereal. What is the function of vitamin D? **[1]**

- ☐ A) It is needed to make haemoglobin for red blood cells.
- ☐ B) It is the main store of energy in the body.
- ☐ C) It helps the body take in calcium for strong bones and teeth.
- ☐ D) It is used to build new muscle tissue for growth.

5 In hospital, patients with lung disease are sometimes given extra oxygen through a mask. Their cells use the oxygen for aerobic respiration. **[1]**

(a) Complete the word equation for aerobic respiration.
glucose + oxygen → _____ + water

(b) In which part of a cell does aerobic respiration take place? **[1]**

- ☐ A) the nucleus
- ☐ B) the cell membrane
- ☐ C) the cell wall
- ☐ D) the mitochondria

(c) Aerobic respiration gives a **controlled** release of energy. Explain what "controlled" means here. **[1]**

6 Lake Nasser is a large lake behind the Aswan High Dam. Its ecosystem includes open water, the muddy lake bed, reed beds near the shore and small rocky islands. **[1]**

(a) Which of these is a **habitat** within the Lake Nasser ecosystem?

- ☐ A) the reed beds near the shore
- ☐ B) all the living things in the lake
- ☐ C) the Nile crocodile
- ☐ D) the amount of sunlight that reaches the water

(b) Water hyacinth is a plant that came from South America. It now grows quickly on parts of the River Nile and covers the surface of the water. Describe **two** ways it can affect other organisms in the ecosystem. **[2]**

7 Pesticide from farms near Lake Nasser washes into the water. Scientists measured the concentration of the pesticide in organisms in one food chain (example data). **[1]**

Organism	Concentration of pesticide in parts per million (ppm)
water plants	0.04
small fish	0.6
Nile perch	3
fish eagle	30

(a) How many times greater is the concentration of pesticide in the fish eagle than in the small fish?

Working

Answer ____ times

(b) Explain why the fish eagle has the highest concentration of pesticide. **[2]**

8 Solar panels are made from silicon. An atom of silicon has 14 protons, 14 neutrons and 14 electrons. **[1]**

(a) Complete the sentence.
Protons have a _____ charge, but neutrons have no charge.

(b) Which sentence describes the **Rutherford model** of the atom? **[1]**

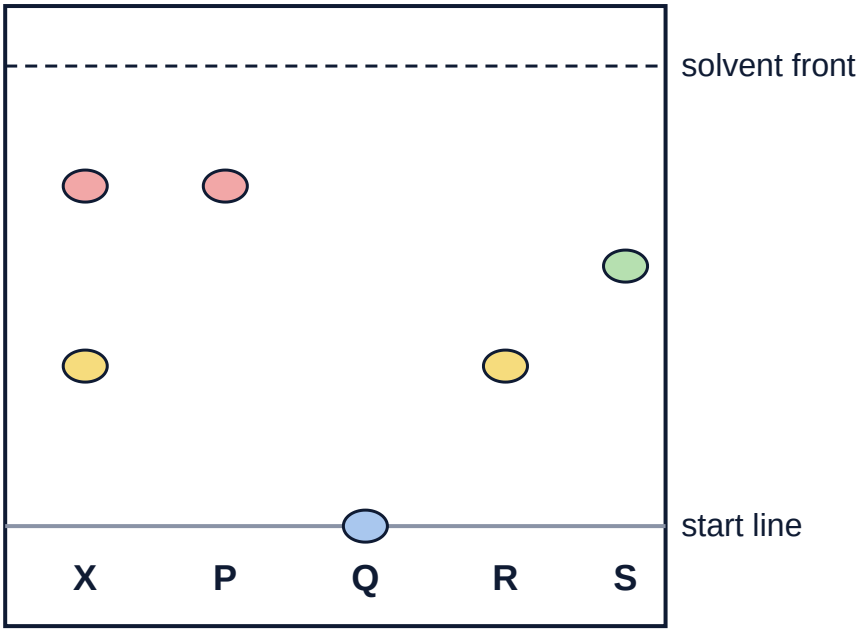
☐ A) A ball of positive charge with negative electrons spread all through it.
☐ B) A solid ball with no charge and no smaller parts inside it.
☐ C) A tiny positive nucleus in the centre, with negative electrons around it and mostly empty space.
☐ D) A large negative nucleus with positive electrons moving around it.

(c) A teacher says: "If an atom were as big as a football stadium, its nucleus would be about the size of a pea on the centre spot." What does this analogy show? **[1]**

☐ A) Atoms are as large as football stadiums.
☐ B) The nucleus is very small compared with the whole atom, so most of the atom is empty space.
☐ C) The nucleus is round and green like a pea.
☐ D) Electrons are much bigger than the nucleus.

(d) Explain what holds the electrons in the atom. **[1]**

9 In the food factory's laboratory, a scientist tests the colouring in a sweet, X. She compares it with four food colours, P, Q, R and S, that are allowed in food. The diagram shows her chromatogram. **[1]**



(a) Which **two** food colours are in sweet X?

(b) Food colour Q did **not** move up the paper. Suggest why. **[1]**

(c) Why must the start line be **above** the level of the solvent in the beaker? **[1]**

- ☐ A) So that the solvent moves up the paper faster.
- ☐ B) So that the paper does not get wet.
- ☐ C) So that the pencil line does not move.
- ☐ D) So that the colours do not dissolve straight into the solvent in the beaker.

10 (continued) **[2]**

The factory also buys salt. A 50 g sample of the salt contains 48 g of sodium chloride. The rest is other substances.

(d) Calculate the percentage purity of the salt. Show your working.

Working

Answer ____ %

11 Omar is a diver at a swimming pool. His weight is 720 N. **[2]**

(a) Omar stands on the side of the pool. The total area of his feet touching the floor is 0.04 m^2. Calculate the pressure on the floor. Show your working.

Working

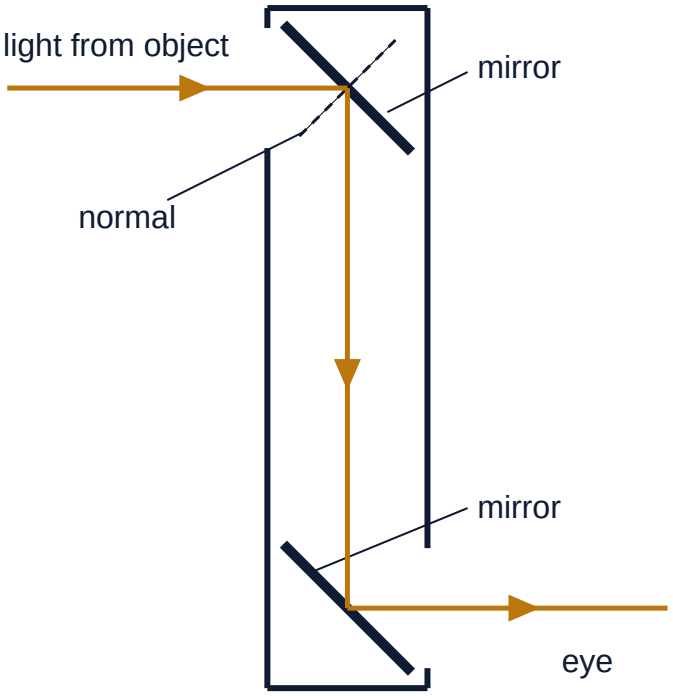
Answer ____ N/m^2

(b) Omar puts on his diving flippers and stands on the floor again. Which statement is correct? **[1]**

- ☐ A) The pressure increases because his weight is bigger.
- ☐ B) The pressure stays the same because his weight has not changed.
- ☐ C) The pressure decreases because the same force acts on a bigger area.
- ☐ D) The pressure decreases because his weight decreases.

(c) Omar's tank of air is left in hot sunshine before a dive. The pressure of the air inside the tank increases. Use ideas about particles to explain why. **[2]**

12 Laila makes a periscope so that she can see over the wall of the swimming pool. It has two plane mirrors. Each mirror is at 45° to the sides of the tube. The diagram shows a ray of light passing through it. **[1]**



(a) What is the angle of incidence at the top mirror?

Working

Answer °

13 (continued) **[1]**

In a different mirror, a ray of light hits the mirror surface. The angle between the ray and the **mirror surface** is 35°.

(b) What is the angle of reflection?

Working

Answer °

14 Laila makes a periscope so that she can see over the wall of the swimming pool. It has two plane mirrors. Each mirror is at 45° to the sides of the tube. The diagram shows a ray of light passing through it. **[1]**

(c) State the law of reflection.

15 The food factory uses an electromagnet above a conveyor belt to pull small pieces of iron out of the cereal. Engineers make a model electromagnet and investigate how the **current** affects the **mass of iron filings** it can lift. They do each test three times. **[1]**

Current in A	Mass lifted (1st) in g	Mass lifted (2nd) in g	Mass lifted (3rd) in g
0.5	4.1	3.9	4.0
1.0	8.2	7.9	8.0
1.5	12.1	11.8	12.1
2.0	16.0	11.2	16.3
2.5	19.8	20.1	20.0

(a) Which equipment do the engineers need to make the electromagnet?

- ☐ A) an iron nail, a coil of insulated copper wire and a power supply
- ☐ B) a plastic rod, bare copper wire and a thermometer
- ☐ C) a bar magnet, a glass rod and a power supply
- ☐ D) an iron nail, a rubber band and a compass

(b) Name the **independent** variable in this investigation. **[1]**

(c) At which current is there an anomalous result? **[1]**

Working

Answer ____ A

(d) Calculate the mean mass lifted at 2.0 A. Do **not** include the anomalous result. **[1]**

Working

Answer g

(e) Describe the relationship between the current and the mass of iron filings lifted. **[1]**

(f) Explain why the factory uses an electromagnet and not a permanent magnet to remove the iron. **[1]**

Paper 2 — 45 minutes — 50 marks

1 A hospital first-aid kit has an instant cold pack and a hand warmer. When the cold pack is squeezed, a solid inside it dissolves in water. A nurse measures the temperature of the cold pack. **[1]**

Time in minutes	0	1	2	3
Temperature in °C	22	12	7	6

(a) By how much does the temperature of the cold pack fall in 3 minutes?

Working

Answer ____ °C

(b) Which word describes the process in the cold pack? **[1]**

- ☐ A) endothermic
- ☐ B) exothermic
- ☐ C) combustion
- ☐ D) neutralisation

2 (continued) **[1]**

The hand warmer contains iron powder. When the packet is opened, the iron reacts with oxygen in the air to make iron oxide, and the packet gets warm.

(c) Write the word equation for this reaction.

(d) How does the temperature change show that the reaction in the hand warmer is **exothermic**? **[1]**

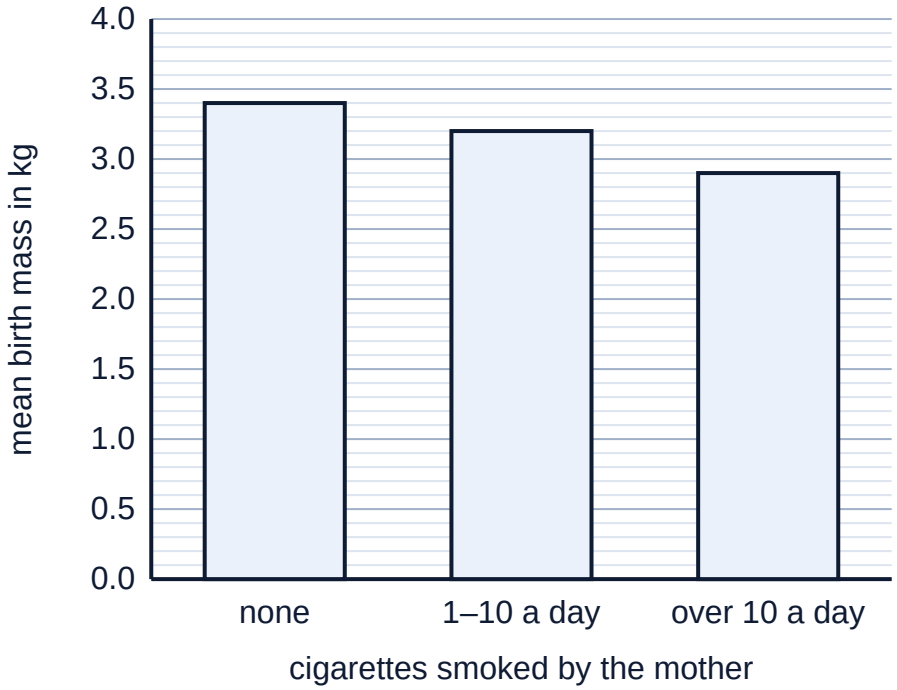
3 A hospital runs a healthy-lifestyle clinic for parents. **[1]**

(a) Animals eat food to get energy and nutrients. Which **two** nutrients can the human body store as an energy store?

- ☐ A) vitamins and minerals
- ☐ B) water and protein
- ☐ C) calcium and iron
- ☐ D) carbohydrates and fats

4 (continued) **[1]**

A hospital study recorded the mean birth mass of babies born to three groups of mothers. The bar chart shows the results (example data).



(b) What was the mean birth mass of babies whose mothers smoked **more than 10** cigarettes a day?

Working

Answer ____ kg

(c) Describe the pattern shown by the bar chart. **[1]**

(d) Which substance in tobacco smoke makes people **addicted** to smoking? **[1]**

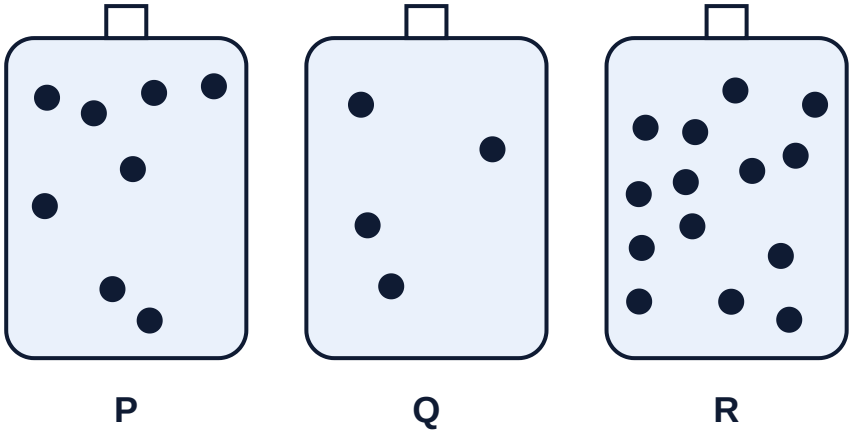
- ☐ A) tar
- ☐ B) carbon monoxide
- ☐ C) nicotine
- ☐ D) oxygen

5 (continued) **[1]**

A leaflet paid for by a tobacco company says: "Smoking a few cigarettes a day does no harm to your health."

(e) Explain why this source of information may be **biased**.

6 In hospital, some patients are given a salt solution through a drip. The diagrams show three drip bags, P, Q and R. Each bag holds the same volume of solution. Each dot represents a number of salt particles. **[1]**



(a) Which bag holds the **most concentrated** solution?

7 (continued) **[1]**

A drip solution is made by dissolving 9 g of salt in water to make 1000 cm^3 of solution.

(b) What mass of salt is in 250 cm^3 of this solution?

Working

Answer g

8 In hospital, some patients are given a salt solution through a drip. The diagrams show three drip bags, P, Q and R. Each bag holds the same volume of solution. Each dot represents a number of salt particles. **[1]**

(c) A nurse adds pure water to bag P. Use ideas about particles to explain why the solution becomes **less** concentrated.

9 A food factory uses two salts: sodium chloride (table salt) and potassium nitrate (a preservative used in some meat products). The table shows the mass of each salt that dissolves in 100 g of water at different temperatures. **[1]**

Temperature in °C	0	20	40	60	80
sodium chloride in g	36	36	37	37	38
potassium nitrate in g	13	32	64	110	169

(a) What mass of potassium nitrate dissolves in 100 g of water at 40 °C?

Working

Answer ____ g

(b) A worker dissolves as much potassium nitrate as possible in 100 g of water at 60 °C. She then cools the solution to 20 °C. Calculate the mass of potassium nitrate crystals that form. Show your working. **[2]**

Working

Answer ____ g

(c) Which statement is correct? **[1]**

- ☐ A) The solubility of both salts increases a lot as the temperature rises.
- ☐ B) The solubility of sodium chloride changes very little as the temperature rises.
- ☐ C) Sodium chloride is more soluble than potassium nitrate at every temperature.
- ☐ D) The solubility of potassium nitrate decreases as the temperature rises.

10 (continued) **[1]**

When the factory's supplier makes potassium nitrate by a chemical reaction, the product is often impure.

(d) Give **one** reason why a reaction may not produce a single pure product.

11 On a car journey, Mona's father explains why different metals are used in the car. The car body is made of steel (mostly iron) coated with a thin layer of zinc. **[1]**

(a) Which list puts the metals in order of reactivity, **most reactive first**?

- ☐ A) magnesium, zinc, iron, copper
- ☐ B) zinc, magnesium, copper, iron
- ☐ C) copper, iron, zinc, magnesium
- ☐ D) iron, zinc, magnesium, copper

(b) Complete the word equation for the reaction of zinc with oxygen. **[1]**

zinc + oxygen → _____

12 (continued) **[1]**

Some electrical connectors in the car are coated with gold. Gold does not react with air, water or dilute acids, even after many years.

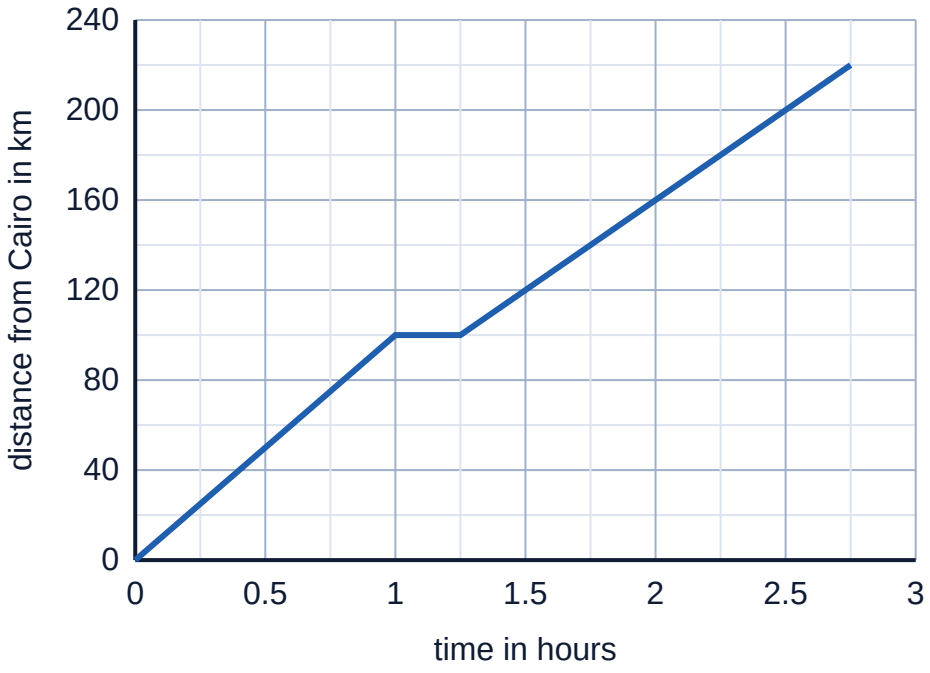
(c) Give the word that describes a substance that is generally unreactive.

13 (continued) **[1]**

Sodium is far too reactive to use in a car.

(d) Describe **one** thing you would see if a small piece of sodium were put into a trough of water.

14 Mona's family drives from Cairo along the Cairo–Alexandria Desert Road. They stop once at a rest house. The distance–time graph shows their journey. **[1]**



(a) Calculate the speed of the car during the **first hour**.

Working

Answer ____ km/h

(b) The family reaches Alexandria after 2.75 hours. Calculate the **average speed** for the whole journey, including the stop. Show your working. **[2]**

Working

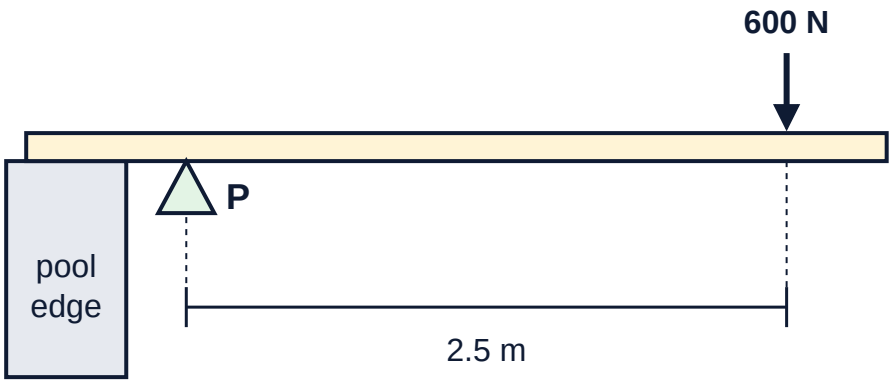
Answer ____ km/h

(c) On a straight part of the road, the car moves at a **constant speed**. Which statement about the forces on the car is correct? **[1]**

- ☐ A) The driving force is bigger than the friction and air resistance.
- ☐ B) The driving force is smaller than the friction and air resistance.
- ☐ C) The forces are balanced: the driving force equals the friction and air resistance.
- ☐ D) There are no forces acting on the car.

(d) Near Alexandria the driver presses the brakes and the car slows down. Explain why, in terms of forces. **[1]**

15 Youssef stands on the end of a diving board at the swimming pool. His weight is 600 N. The diagram shows a model of the board. The board can turn about point P. **[2]**



(a) Calculate the moment of Youssef's weight about P. Show your working.

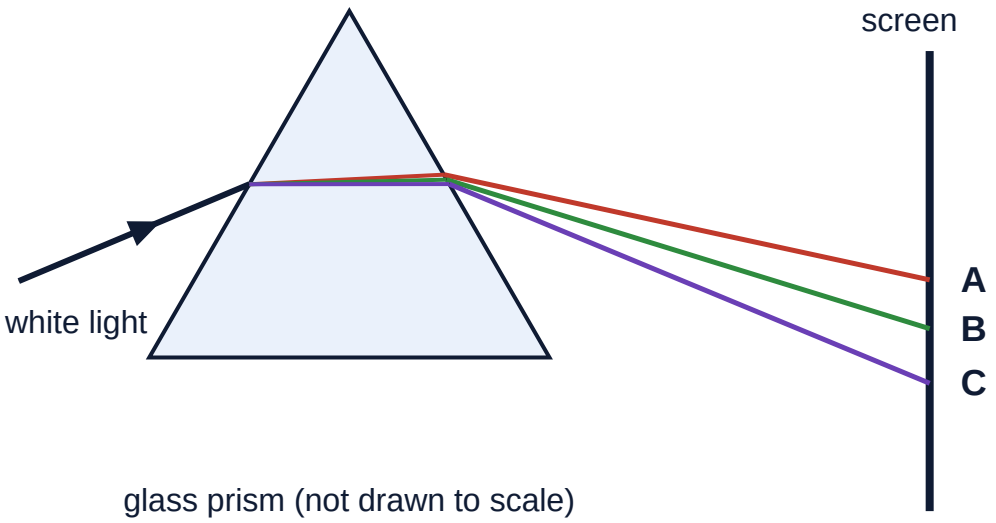
Working

Answer ____ N m

(b) Youssef walks along the board **towards P**. What happens to the moment of his weight about P? **[1]**

- ☐ A) It increases.
- ☐ B) It stays the same.
- ☐ C) It changes direction.
- ☐ D) It decreases.

16 Amira is a photographer. She shines a narrow beam of white light through a glass prism onto a screen and photographs the colours. **[1]**



(a) Which letter, A, B or C, shows where **violet** light reaches the screen?

(b) Explain why light changes direction when it enters the glass prism. **[1]**

17 (continued) **[1]**

Amira photographs a red apple with a green leaf on it. She puts a red filter in front of her camera lens.

(c) How do the apple and the leaf look in her photograph?

- ☐ A) The apple looks red and the leaf looks black.
- ☐ B) The apple looks red and the leaf looks green.
- ☐ C) The apple looks black and the leaf looks green.
- ☐ D) The apple and the leaf both look red.

18 In the food factory's laboratory, Karim investigates how the temperature of water affects diffusion. He adds one drop of food colouring to a beaker of still water and times how long the colour takes to spread evenly through the water. **[1]**

Temperature of water in °C	**20**	**30**	**40**	**50**	**60**
Time to spread evenly in s	330	250	290	130	90

(a) Which hypothesis can Karim test with this investigation?

- ☐ A) Food colouring tastes better in warm water.
- ☐ B) Particles of food colouring spread out faster in warmer water.
- ☐ C) Food colouring is made of particles that are too small to see.
- ☐ D) Diffusion is the most important process in the factory.

(b) What is the **dependent** variable? **[1]**

- ☐ A) the temperature of the water
- ☐ B) the volume of water in the beaker
- ☐ C) the size of the drop of food colouring
- ☐ D) the time for the colour to spread evenly

(c) At which temperature is the result anomalous? **[1]**

Working

Answer ____ °C

(d) Describe the trend in the results and explain it using ideas about particles. **[2]**

(e) The water at 60 °C is a hazard. Describe the risk and **one** way to reduce it. **[1]**

19 A large solar farm near Aswan uses thousands of solar panels to generate electricity. **[1]**

(a) Which list contains **only** renewable energy resources?

- ☐ A) solar, coal, wind
- ☐ B) solar, wind, tidal
- ☐ C) natural gas, oil, coal
- ☐ D) tidal, oil, solar

(b) Explain **two** advantages of generating electricity from solar power near Aswan instead of by burning natural gas. **[2]**

(c) Some of the packaging for the panels is made of **bioplastic**. What is a bioplastic? **[1]**

- ☐ A) a plastic made only from crude oil
- ☐ B) a metal that can bend like plastic
- ☐ C) a plastic that can never break down
- ☐ D) a plastic made from plant materials such as corn starch or sugar cane

20 Engineers at the solar farm study data about Aswan's weather and climate. **[1]**

(a) Which sentence describes Aswan's **climate**, not its weather?

- ☐ A) It was 41 °C in Aswan yesterday afternoon.
- ☐ B) There is a strong wind in Aswan this morning.
- ☐ C) Aswan is hot and very dry, with almost no rain in most years.
- ☐ D) It might rain in Aswan tomorrow.

(b) Scientists study tiny bubbles of air trapped in very old ice in Antarctica. What does this evidence show about the Earth's climate over the last 800 000 years? **[1]**

- ☐ A) It has changed many times between cold ice ages and warmer periods.
- ☐ B) It has stayed at the same temperature all the time.
- ☐ C) It has become warmer every single year.
- ☐ D) There was only one ice age, and it ended last century.

(c) Explain how burning fossil fuels can change the Earth's climate. **[2]**

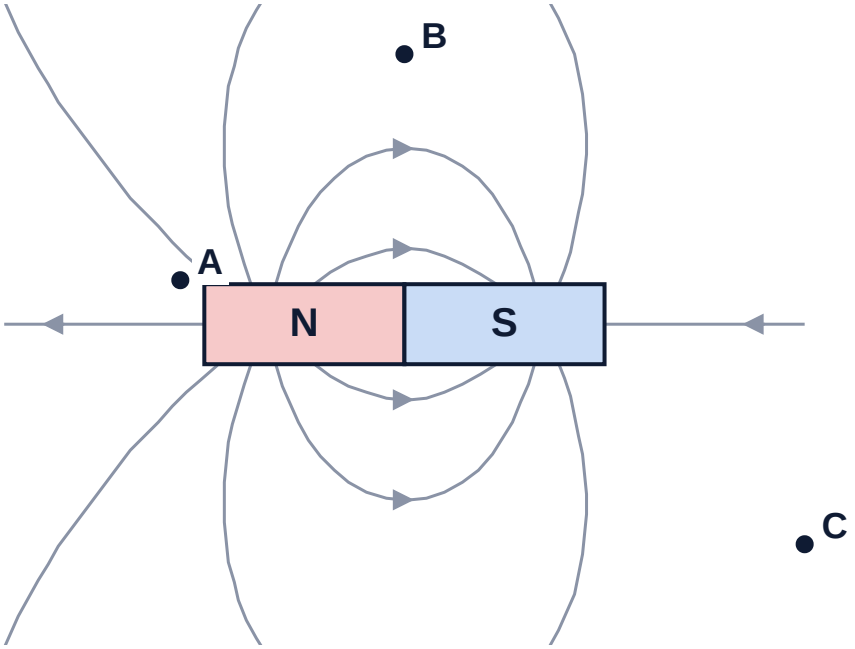
21 Aswan has very clear night skies. A school astronomy club looks at the Milky Way. **[1]**

(a) Which describes a **galaxy**?

- ☐ A) one star with planets orbiting it
- ☐ B) a huge group of stars and planetary systems, with dust and gas, held together by gravity
- ☐ C) a cloud of gas around one planet
- ☐ D) a large rock that orbits the Sun

(b) Describe how asteroids formed. **[1]**

22 The diagram shows the magnetic field around a bar magnet. A, B and C are three points in the field. **[1]**



(a) At which point, A, B or C, is the magnetic field **strongest**?

23 (continued) **[1]**

On the drive to Alexandria, Mona uses a compass to find north.

(b) Why does the Earth have a magnetic field?

- ☐ A) Sunlight magnetises the Earth's surface.
- ☐ B) The Moon pulls on the compass needle.
- ☐ C) The Earth's core acts like a magnet.
- ☐ D) The Earth's crust is made of bar magnets.